breadth of engineering or the scalability in engineering. They ended up investing more and more capital in order to survive. Business is only two things, it is capital and it is labour. In our case we have talented labour. However, we never put too much stress in capital because we were able to create the world's IT industry here in India. The other countries did not have such talented labour and hence they had no choice but to put in a lot of investment."

China is the biggest example of what can happen when you put in a lot of capital. The country, some decades back, lacked quality manpower and hence it grew by investing capital. Semiconductors dot Org forecasts that the Chinese will be investing anything around \$12 to \$15 billion. Thanks to the government of China the country has announced over 110 fab projects since the year 2014. The investments in these fabs are around \$196 billion.

"The good they did was they did not allow the foreign companies to come in and dominate. They had home-grown companies. That is why there is no Google and Facebook in China. They never hesitated in putting barriers on the chip-side too. China, in terms of chips, probably has every catalogue product being done by someone in China. There are hundreds of companies like that and then there are hundreds of fabs who can support these companies," shares Raja Manickam.

Why China has been so successful in the same, as Manickam answers, is because companies there do not want to do everything. One company focuses on one thing, which could be design, support, or fabrication. The semiconductor industry has many segments where the small and medium sized companies can help. Equipment required for semiconductors is also a big segment.

"At the same time there are things that SMEs can make for the big instrument making brands. If we can incentivise big names like Intel and TSMC, we can also ask them what they can do for the SMEs to prosper and add new things to their product range. We might want to be more savvy about the PLI schemes and ask the benefitters what are they giving us back apart from jobs," shares Manickam.

Semiconductor Ecosystem - The Myths

- **Fabs can only be successful in countries where there is a huge demand for semiconductors**
  "It really is a myth. If you look at where the semiconductors are manufactured, whether it is fab or post-fab OSATs, they have got nothing to do with the market. The world's largest market for consumption of semiconductors is in the US. Yet, there is hardly any manufacturing in the US. The government of the US is now trying to offer incentives for setting up fabs due to strategic reasons," shares Raja Manickam.
- **It is critical to have a fab in the country to do anything backend around semiconductors**
  "Wafers that are fabricated in Taiwan are not necessarily packaged in Taiwan. Semiconductor fabrication companies in Taiwan ship a lot of wafers to the likes of Malaysia, Singapore, Philippines and many other parts of the world for packaging and testing," shares Raja Manickam.
- **There is a huge investment required**
  "It is probably true to some extent, but a lot depends on which segment you are going to play into. Whether you take the number of wafers that come out or you take it in terms of packaged parts, the majority of the units do not require 5, 7, or 3nm types of fabs. Majority is probably in the 28nm and above segment. There is plenty of business here," shares Manickam.

The opportunity(ies)

The semiconductor ecosystem is not just about designing, fabricating, or packaging semiconductors. It is actually a Pandora's box of opportunities for small and medium companies. The opportunity to supply raw materials, the opportunity for making software, the opportunity for transporting chips, and more. The biggest of these opportunities lies in working consistently to create dominance in what a company, whether small or big, does around semiconductors.

"There is a big number of small companies playing in this arena. They are supplying chemicals, raw materials, products like valves, and much more to the ecosystem. There is whole bunch of requirements. Small and medium companies need to pick where their strength lies. For example, if a founder has a major in chemistry, there is enough room for working around the chemistry used in the field of semiconductors," shares Manickam.

Applied Materials has recently applied to the Central government and the government of Karnataka to be an anchor account that can create ancillaries for its procurement requirements. The company, on an average, buys \$3 billion worth of specialised equipment and parts from Asia Pacific region. The company has expressed that India can supply materials to it worth over ₹10 to 20 billion annually.

"We have discovered gems in India during this process. These belong to the verticals of specialised assembly components, and some of these are also SPECS applicants. There will also be a supply chain for equipment that we will generate out of India," says Ashwini.

The fact that there are a lot of Indians working in the semiconductor ecosystem outside India can also prove to be one of the biggest strengths for the country. When Morris Chen, the founder of TSMC, started working in the semiconductor fab business, there were only a handful of Taiwanese nationals in the vertical. There are a lot more Indians working in the semiconductor ecosystem than what was the scene for Taiwan in the 1980's.

"People like me went outside India in the 1970's and 80's. A lot of us still carry the baggage of what India was like at that time. Now there's a lot more pull to bring in stuff and technology than the push," says Manickam. EFY

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